

OpenCPU SDK

FTP API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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Contents

1 Overview	5
2 Data Structure	6
2.1 nwy_ftp_handle_t	6
2.2 nwy_ftp_put_mode_e	6
2.3 nwy_ftp_param_t	6
2.4 nwy_ssl_ftp_type_e	6
2.5 nwy_ftp_ssl_cfg_t	7
2.6 FTP Event nwy_ftp_event_e	7
2.7 Callback Function Definition	8
3 APIs	9
3.1 nwy_ftp_login	9
3.2 nwy_ftp_logout	9
3.3 nwy_ftp_get	10
3.4 nwy_ftp_put	10
3.5 nwy_ftp_get_filesize	11
3.6 nwy_ftp_del_file	11

About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Gao Zheng

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the FTP-related APIs in the OpenCPU SDK, specifically for FTP/SSL FTP operations.

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2 Data Structure

2.1 nwy_ftp_handle_t

```
typedef void *nwy_ftp_handle_t;
```

2.2 nwy_ftp_put_mode_e

```
typedef enum
{
    FTP_STOR_MODE = 1, // Creates a file on the server for writing data. If the file exists,
                        // it overwrites the original file.
    FTP_APPE_MODE,    // Creates a file on the server for writing data. If the file exists,
                        // data is appended to the end of the file.
}nwy_ftp_put_mode_e;
```

2.3 nwy_ftp_param_t

```
typedef struct{
    unsigned short cid;
    int timeout_s;
    char *host;
    int port;
    char *username;
    char *password;
    unsigned char mode;//0 pasv 1:port
    nwy_ftp_resultcb cb;//callback function
}nwy_ftp_param_t;
```

2.4 nwy_ssl_ftp_type_e

```
typedef enum
{
    SSLFTP_TYPE_EXPLICIT =1,//explicit SSL
    SSLFTP_TYPE_IMPLICIT //implicit SSL
}nwy_ssl_ftp_type_e;
```

2.5 nwy_ftp_ssl_cfg_t

```
typedef struct{
    nwy_ssl_ftp_type_e ftps_type;
    nwy_ssl_conf_t ssl_config;
}nwy_ftp_ssl_cfg_t;
```

2.6 FTP Event nwy_ftp_event_e

```
typedef enum
{
    NWY_FTP_EVENT_DNS_ERR,
    NWY_FTP_EVENT_OPEN_FAIL,
    NWY_FTP_EVENT_CLOSED,
    NWY_FTP_EVENT_LOGOUT,
    NWY_FTP_EVENT_DATA_GET,
    NWY_FTP_EVENT_DATA_GET_ERROR,
    NWY_FTP_EVENT_DATA_OPEND,
    NWY_FTP_EVENT_DATA_CLOSED,
    NWY_FTP_EVENT_DATA_SETUP_ERROR,
    NWY_FTP_EVENT_DATA_GET_FINISHED,
    NWY_FTP_EVENT_LOGIN,
    NWY_FTP_EVENT_SIZE,
    NWY_FTP_EVENT_PASS_ERROR,
    NWY_FTP_EVENT_FILE_NOT_FOUND,
    NWY_FTP_EVENT_FILE_SIZE_ERROR,
    NWY_FTP_EVENT_FILE_DELE_SUCCESS,
    NWY_FTP_EVENT_FILE_DELE_ERROR,
    NWY_FTP_EVENT_DATA_PUT_FINISHED,
    NWY_FTP_EVENT_DATA_PUT_ERROR,
    NWY_FTP_EVENT_CLOSED_SSL_FAIL,
    NWY_FTP_EVENT_SSL_CONNECT_FAIL,
    NWY_FTP_EVENT_SSL_CONNECTED,
    NWY_FTP_EVENT_UNKOWN
}nwy_ftp_event_e;
```

Event id	Note
NWY_FTP_EVENT_DNS_ERR,	DNS resolution error
NWY_FTP_EVENT_OPEN_FAIL	Control link socket open failure
NWY_FTP_EVENT_CLOSED	Control link socket closed
NWY_FTP_EVENT_LOGOUT	FTP logout
NWY_FTP_EVENT_DATA_GET	Received FTP GET data
NWY_FTP_EVENT_DATA_GET_ERROR	FTP GET error
NWY_FTP_EVENT_DATA_OPEND	FTP data link opened successfully
NWY_FTP_EVENT_DATA_CLOSED	FTP data link closed

NWY_FTP_EVENT_DATA_SETUP_ERROR	FTP data link open failure
NWY_FTP_EVENT_DATA_GET_FINISHED	FTP GET completed
NWY_FTP_EVENT_LOGIN	FTP login successful
NWY_FTP_EVENT_SIZE	FTP file size retrieved
NWY_FTP_EVENT_PASS_ERROR	FTP password error
NWY_FTP_EVENT_FILE_NOT_FOUND	File not found on server
NWY_FTP_EVENT_FILE_SIZE_ERROR	FTP file size retrieval failed
NWY_FTP_EVENT_FILE_DELE_SUCCESS	File successfully deleted on server
NWY_FTP_EVENT_DATA_PUT_FINISHE	FTP file upload completed
NWY_FTP_EVENT_DATA_PUT_ERROR	FTP PUT error
NWY_FTP_EVENT_CLOSED_SSL_FAIL	SSL close failure
NWY_FTP_EVENT_SSL_CONNECT_FAIL	SSL connection failure
NWY_FTP_EVENT_SSL_CONNECTED	SSL connection successful
NWY_FTP_EVENT_UNKOWN	Internal SSL error

2.7 Callback Function Definition

```
typedef struct{
    unsigned short channel; //data channel
    nwy_ftp_handle_t ftp_handle; //FTP client handle
    unsigned int event; /* Event id */
    unsigned int data_len; /* Get the length of the received data */
    char* data; /* Acquired data content */
}nwy_ftp_result_t;
typedef void (*nwy_ftp_resultcb)(nwy_ftp_result_t *event);
```


3 APIs

The interface function definitions are located in `nwy_ftp_api.h` for FTP-related operations.

For encrypted FTP operations, the SSL configuration file (`nwy_ssl_config.h`) must be utilized.

3.1 `nwy_ftp_login`

Format	<code>nwy_ftp_handle_t nwy_ftp_login(nwy_ftp_param_t *param , nwy_ftp_ssl_cfg_t *sslcfg)</code>
Description	Login to an FTP/SSLFTP server.
Parameters	<code>nwy_ftp_param_t *param</code> : FTP parameter unsigned short channel: Data channel (1~7). Each channel supports one FTP client. int timeout_s: Timeout char *host: Server address or domain name. int port: Server port. char *username: Login username. char *password: Login password. unsigned char mode: 0 for passive mode, 1 for active mode. nwy_ftp_resultcb cb: FTP status callback function. <code>nwy_ftp_ssl_cfg_t *sslcfg</code> : SSL FTP configuration parameters: Refer to the header file <code>nwy_ssl_config.h</code> for SSL configuration details. Setting the parameter to NULL enables an unencrypted mode.
Return Value	FTP client handle Success: non-NULL Failure: NULL
Remarks	Supports up to two FTP clients online simultaneously.

3.2 `nwy_ftp_logout`

Format	<code>nwy_error_e nwy_ftp_logout (nwy_ftp_handle_t handle);</code>
Description	Logout from an FTP/SSLFTP connection.
Parameters	handle: FTP/SSLFTP client handle.
Return Value	Check <code>nwy_error_e</code>

3.3 nwy_ftp_get

Format	nwy_error_e nwy_ftp_get(nwy_ftp_handle_t handle, nwy_ftp_get_param_t *ftp_get_param)
Description	Download a file from the FTP/SSLFTP server.
Parameters	nwy_ftp_handle_t handle: FTP/SSLFTP client handle. <pre>typedef struct{ char * remotefile; unsigned char type; int offset; int len; }nwy_ftp_get_param_t;</pre> Structure parameter type: char * remotefile: File name to retrieve. unsigned char type: 1. ASCII 2. binary int offset: File offset (0 for full download) int len: File length (0 for full download)
Return Value	Check nwy_error_e
Remarks	Use nwy_ftp_get_filesize and event NWY_FTP_EVENT_SIZE to get the total file size.

3.4 nwy_ftp_put

Format	nwy_error_e nwy_ftp_put(nwy_ftp_handle_t handle, nwy_ftp_put_param_t *ftp_put_param);
Description	Upload data to a specified file on the server.
Parameter	nwy_ftp_handle_t handle: FTP/SSLFTP client handle. Structure parameter type: <pre>typedef struct{ unsigned char eof; char * remotefile; unsigned char *data; int len; unsigned char type; nwy_ftp_put_mode_e mode; }nwy_ftp_put_param_t;</pre> unsigned char eof: Whether the data upload has ended. 0x00 (not finished), 0x01 (finished).

	char *remotefile: Server file name to upload data to. unsigned char *data: Data to write. int len: Data length (max 8096 bytes). unsigned char type: 1 for ASCII, 2 for binary. nwy_ftp_put_mode_e mode: 1. Creates a file on the server for writing data. If the file exists, it overwrites the original file. 2. Creates a file on the server for writing data. If the file exists, data is appended to the end of the file.
Return Value	Check nwy_error_e

3.5 nwy_ftp_get_filesize

Format	nwy_error_e nwy_ftp_get_filesize(nwy_ftp_handle_t handle, const char* filename);
Description	Gets the length of a file on the server.
Parameters	nwy_ftp_handle_t handle: FTP/SSLFTP client handle. const char* remotefile: File name
Return Value	Check nwy_error_e

3.6 nwy_ftp_del_file

Format	nwy_error_e nwy_ftp_del_file(nwy_ftp_handle_t handle, const char *remotefile);
Description	Delete that file on the server
Parameters	handle: FTP/SSLFTP client handle. remotefile: File name to delete.
Return Value	Check nwy_error_e